

Denis Bolonin

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UK full-time work permit, valid until Jan 2028

London, UK

PROFESSIONAL SUMMARY

Software Engineer with **over 5 years** of experience building high-throughput data pipelines and backend systems across fintech and SaaS environments. Proven track record of automating manual data workflows end-to-end - from ingestion and validation to structured storage and reconciliation. Applied research background in large-scale model training pipelines (PyTorch, multimodal datasets) through two independent research projects.

Comfortable owning infrastructure across the full stack: databases, streaming, orchestration, and cloud deployment. Actively engaged in the open-source community - contributor to ClickHouse, a prominent open-source DBMS, with technical work recognized by the project's CTO.

WORK EXPERIENCE

Senior Software Engineer (Data Engineering & Backend)

08/2024 - Present

[digital.sfpe](#), a software development SaaS startup

- Developed from the ground up a comprehensive software solution tailored for fire protection engineering companies to ensure regulatory compliance. This solution automates critical processes such as contract management, data validation, format control, minimizes errors and proactively monitors deadlines. The implementation of this software significantly improved efficiency, enabling a **single person** to manage the workload that previously required **3-4 full-time employees**.
- Implemented autofill process for EMERCOM contract notifications, reducing the processing time per notification from 8-10 minutes to under 1.5-2 minutes (**over 80% speed up**).
- Engineered an end-to-end subscription billing data pipeline - order intake, **OCR-based payment document parsing**, invoicing, and reconciliation - replacing an almost fully manual workflow and eliminating human error across the billing pipeline.

Software Developer

07/2022 - 07/2024, 2 years 1 month

[Sber](#), one of the biggest fintech companies

- Developed a utility for financial models for a high-value investment project, specifically valued at 2 billion USD, **accelerating** model calculations **by over 50%** through concurrent processing.
- Engineered an automated data pipeline for depositary settlements, handling **ingestion, validation, and reconciliation** and fully eliminating manual intervention.
- Developed a team productivity dashboard for Scrum Masters, enabling data-driven decision-making and **increasing team throughput by 15%**.

Open Source Developer

10/2020 - 06/2022, 1 year 9 months

[ClickHouse](#), opensource DBMS

- Researched and implemented UDP-based replication protocols (UDT, μ TP/uTP, ENet) for ClickHouse, evaluating their performance and laying groundwork for future QUIC protocol studies.
- Integrated Rust into ClickHouse build system, enabling C++ and Rust interoperability. Implemented and integrated the BLAKE3 hashing function, achieving almost **100% performance increase** with a new hashing function compared to existing functions.
- Implemented YAML configuration support for ClickHouse, providing users with a more readable and convenient configuration format.

TECHNICAL SKILLS

Data Engineering: ETL Pipeline Design, Data Warehousing, Data Modeling, Batch & Stream Processing, Apache Kafka, Apache Spark, Hadoop, RabbitMQ, ZooKeeper, Data Validation & Reconciliation, Data Pipeline Development

Databases & Data Storage: PostgreSQL, ClickHouse, Redis, SQLite

Programming Languages: Python, SQL, C++, Rust, Bash, R, JavaScript, C

Backend Development: Django, FastAPI, Flask, REST APIs, gRPC, Microservices

Cloud & DevOps: Docker, Kubernetes, OpenShift, Jenkins, Helm, AWS, Istio, CI/CD, Infrastructure as Code

Machine Learning (Applied): PyTorch, Fine-Tuning Pipelines, Data Preprocessing for Multimodal ML, OCR

Monitoring & Analytics: Prometheus, Grafana, Kibana, Performance Optimization

Testing: pytest, Unit Testing, Functional Testing, Fuzzing

Project Management: Agile, Scrum, Jira, Confluence, Team Leadership, Mentoring

RESEARCH PROJECTS

Multimodal Semantic Analysis of Scientific Publications

2025 - 2026

Research Project, HSE University

Code: github.com/BoloniniD/Multimodal-Scientific-Publication-Analysis

- Built a data pipeline for extracting and aligning text and figure data from Russian-language scientific publications, powering a multimodal vision-language training pipeline.
- Applied Fisher Information Matrix analysis over model layers to selectively allocate QLoRA adapters to the most task-relevant 40% of layers, cutting the trainable adapter footprint by 60% while outperforming uniform fine-tuning across two downstream tasks.

Automated Layer Selection for Efficient Fine-Tuning of Medical Image Segmentation Models

2024 - 2025

MSc thesis, Computer Science, Queen Mary University of London

Code: github.com/BoloniniD/Automated-Layer-Selection-for-Efficient-Fine-Tuning

- Designed and processed a multi-stage abdominal CT dataset pipeline for sequential model training, handling data validation and stage-wise partitioning for a three-stage learning protocol.
- Built an experimentation framework benchmarking three fine-tuning strategies, optimizing for both storage and compute efficiency to run entirely on a single consumer GPU.

EDUCATION

Master's degree in Computer Science. Queen Mary University of London, London, UK

09/2024 - 09/2025

Software and Data Engineering stream

Bachelor's degree in Computer Science. HSE University, Moscow, Russia

09/2019 - 06/2023

Applied Mathematics and Informatics, Distributed systems

CERTIFICATIONS

ICAgile Certified Professional

12/2022

Completed Agile training, received the ICAgile Certified Professional

[Certificate Link](#)

Applied statistical analysis

05/2022

Completed additional Applied Statistical Analysis course, HSE University

PUBLICATIONS

More Than 2x Faster Hashing in ClickHouse Using Rust

Co-authored technical blog post on integrating Rust into ClickHouse. The work was recognized by ClickHouse CTO Alexey Milovidov, who wrote: "I'm highly impressed by the work of Denis Bolonin that made it possible!" [ClickHouse Official Blog](#)

ACHIEVEMENTS

Winner of the Finance block hackathon at Sber

09/2023

- Developed the backend data layer for a mobile approval platform, including storage and management of user data and incoming approval requests.
- Collaborated with the team on product ideation and workflow design.
- The solution replaced a cumbersome legacy process with a more engaging, gamified approval experience.